

6 Conic Sections

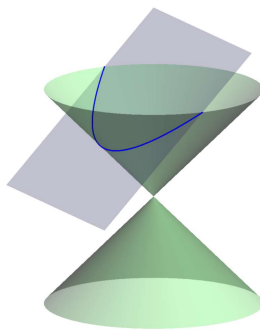
We consider the traditional conic sections and relate their basic properties to modern equations.

Definition 6.1. A (non-degenerate) conic section is the intersection of a right circular cone²² with a plane not containing the vertex. Depending on the orientation of the intersecting plane, we obtain one of the three standard types.

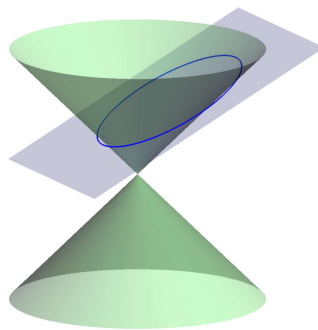
Parabola The plane intersects in a single curve either on the upper or lower half of the cone. Equivalently, the plane is parallel to one of the lines defining the cone.

Ellipse The intersection is a single closed curve.

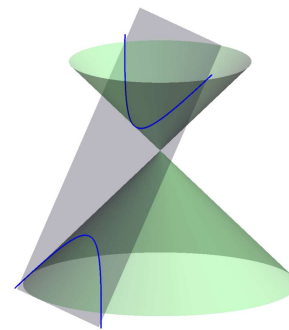
Hyperbola The intersection comprises two curves, one each on the upper and lower halves of the cone.



Parabola



Ellipse



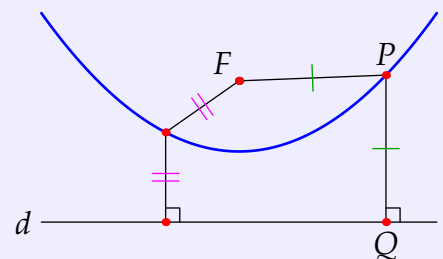
Hyperbola

6.1 The Focus & Directrix of a Parabola

The above conic section definitions are hard to work with because they are three dimensional. We instead make alternative definitions as curves in the plane.

Definition 6.2. Let d be a line and F a point not on that line. The *parabola* (in the plane spanned by d and P) with *directrix* d and *focus* F is the set of points P equidistant from both. As pictured, the parabola is the set of points for which

$$|FP| = |PQ|$$



The definition is strictly *synthetic* since equidistant really means that the line segments $\overline{FP}$ and $\overline{PQ}$ are *congruent*. To interpret this *analytically* we choose axes and co-ordinates.

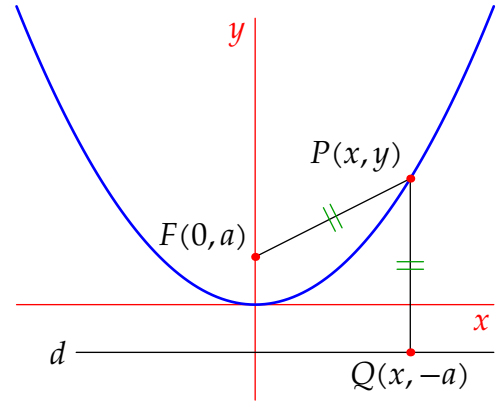
²²A *cone* is a surface obtained by joining all points on a curve with a fixed point, the *vertex*. A right circular cone is when this curve may be chosen to be a *circle* such that the line from its center to the vertex (the *axis*) is perpendicular (*right angles*) to the circle.

Without loss of generality, choose the origin to lie on the parabola halfway between the focus and directrix, with the focus on the positive y -axis. Otherwise said, for some $a > 0$, we have $F = (0, a)$ and the directrix is $y = -a$.

Now we follow the definition: if $P = (x, y)$ is a point on the parabola, then $Q = (x, -a)$, whence

$$\begin{aligned} |FP| = |PQ| &\iff |FP|^2 = |PQ|^2 \\ &\iff x^2 + (y - a)^2 = (y + a)^2 \\ &\iff x^2 + y^2 - 2ay + a^2 = y^2 + 2ay + a^2 \end{aligned}$$

This simplifies to recover a familiar equation:



Theorem 6.3 (Canonical form of a parabola). *Given a parabola, there exist rectangular co-ordinates/axes with respect to which the parabola has equation*

$$x^2 = 4ay$$

where $a > 0$.

Of course, parabolas are often not always presented in canonical co-ordinates.

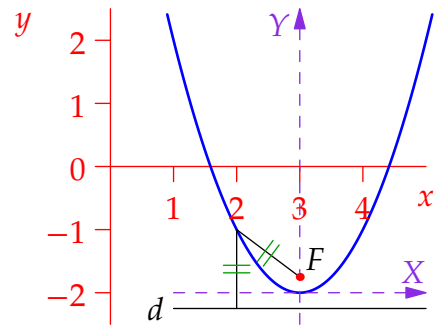
Example 6.4. Find the focus and directrix of the parabola with equation $y = x^2 - 6x + 7$. Start by completing the square!

$$y = x^2 - 6x + 7 = (x - 3)^2 - 2$$

By defining new co-ordinates $X = x - 3$ and $Y = y + 2$, we obtain the canonical form

$$X^2 = Y = 4aY \quad \text{where} \quad a = \frac{1}{4}$$

The focus has co-ordinates $(X, Y) = (0, \frac{1}{4})$ and the directrix is $Y = -\frac{1}{4}$. Translating back to our original co-ordinates (x, y) , we see that the focus and directrix are



$$F = \left(0 + 3, \frac{1}{4} - 2\right) = \left(3, -\frac{7}{4}\right), \quad d : y = -\frac{1}{4} - 2 = -\frac{9}{4}$$

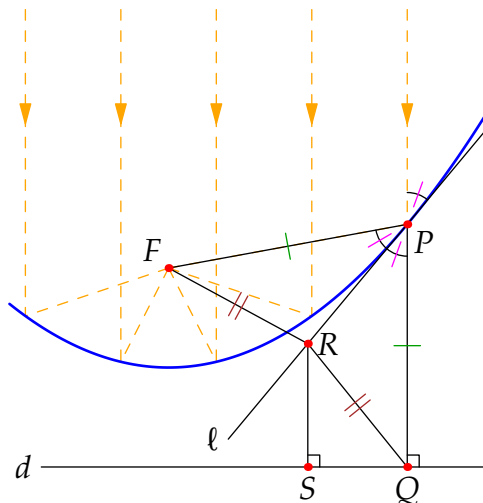
Canonical co-ordinates can also be rotated relative to given axes, but this would take us too far afield. You will likely return to such ideas in an upper-division linear algebra course.

Why call it the *Focus*?

Imagine a *parabolic mirror*: a reflective surface obtained by rotating a parabola around its canonical y -axis. The rays from a light source at ‘vertical infinity’ (e.g., the sun) would arrive parallel to the y -axis. The parabolic mirror will *focus* these rays onto the focus of the original parabola! This is the reason satellite dishes are parabolic, with the receiver placed at the focus.

Convincing ourselves of this requires only a little triangle geometry. Let P lie on the parabola, Q be its nearest point on the directrix, and bisect angle $\angle FPQ$ with the line ℓ . Let $R \neq P$ be any other point on ℓ and drop the perpendicular to d at S .

1. Side-Angle-Side for $\triangle FPR \cong \triangle QPR$ tells us that $|FR| = |RQ|$.
2. Since $\overline{RQ}$ is the hypotenuse of a right-triangle, we see that $|RS| < |RQ| = |FR|$.
3. It follows that all points $R \neq P$ on ℓ lie on the same side of the parabola (closer to d than F), whence ℓ is tangent to the parabola.
4. A vertical ray hitting the parabola at P is reflected so that its angle of incidence equals its angle of reflection. Since ℓ is tangent, this angle equals $\angle QPR \cong \angle FPR$. The ray therefore reflected onto $\overline{PF}$ and passes through F .



Exercises 6.1. Key concepts: *Slicing a cone with a plane, Focus and directrix definition of a parabola, Canonical form by completing the square, Meaning of **focus***

1. Put each parabola in canonical form, and thus find its focus and the equation of its directrix. Sketch each parabola.
 - (a) $x^2 + 2x - y = 0$
 - (b) $4y^2 - 3x + 4y + 1 = 0$
2. A common design for a flashlight places a lightbulb at the focus of a parabolic mirror. Explain why. What would happen to the light beam if the bulb were placed just a little further than the focus from the mirror?
3. A right-circular cone centered on the z -axis has equation $z^2 = x^2 + y^2$ (horizontal cross-sections are circles).
 - (a) Intersect this cone with the plane with equation $z = y + k$ for some constant k . Show that the resulting curve is indeed a parabola.
 - (b) Determine the focus and directrix, and how the constant a in the canonical form relates to k .
(Hint: You'll need to consider a factor of $\sqrt{2}$ with y to compensate for the slope of the intersecting plane—why?)